

Projections of the sardine population using an operating model which does not estimate stock-recruitment relationships during conditioning

C.L. de Moor*

Correspondence email: carryn.demoor@uct.ac.za

The sardine resource is projected for 20 years using the sardine-only 'Scoping' MSE with an operating model which estimated historical annual recruitment to the Cool Temperate (CTS) and Warm Temperate (WTS) stocks independently. Future recruitment is generated in two ways: (a) from the historical median and (b) from a Hockey-Stick 'like' relationship, with the standard deviation calculated non-parametrically. Results with/out annual deviations in future recruitment show the impact this recruitment variability has on the range of future projections.

Keywords: recruitment variability, sardine, stock recruitment, South Africa

Background

There are difficulties in fitting stock recruitment relationships after conditioning the assessments for the South African sardine stocks. In particular, carrying capacity has not been robustly estimated when assuming a Beverton Holt stock recruitment relationship for Warm Temperate sardine (WTS) (de Moor 2025a). For Cool Temperate sardine (CTS), very low estimates of recruitment in some years can result in stock recruitment relationships estimated to be near-zero (de Moor 2025b). This document considers two simple non-parametric 'relationships' for South African sardine.

Methods

The sardine population is projected forward using the Scoping MSE (de Moor 2025c), using an Operating Model which estimated historical recruitment independently, i.e. $R_y^{CTS}/10 \sim U(0,50)$, $R_{c,y}^{WTS}/10 \sim U(0,50)$. Two alternatives are considered for future stock-recruitment 'relationships':

- a) Future recruitment is generated from the median historical (1984-2023) recruitment.
- b) Future recruitment is generated from a relationship intended to mimic the Hockey-Stick, with the hinge point set at the 8th lowest SSB for WTS and the 10th or 20th lowest SSB for CTS¹. Recruitment is assumed to decrease linearly to zero below the hinge point spawner biomass and is assumed to correspond to the median recruitment of points above the hinge point for higher spawner biomass values.

The three pulse years of 2000-2002 were excluded from estimating these 'relationships' for WTS. Future pulse recruitments of WTS are not simulated in this document.

The standard deviation was calculated using a non-parametric method. It was set equal to the residual that corresponds to approximately including 68% of the residuals (Butterworth pers comm.). The absolute residuals, $|\ln(R_y^j) - \ln(Rest_y^j)|$, were ordered and then the 25th/27th out of 37/40 ordered historical years was selected for (a). There are 40 years for CTS and 37 years for WTS as the three pulse years are excluded. For (b), the 20th out of 29 ordered years for WTS and the 20th / 14th out of 30/20

* MARAM (Marine Resource Assessment and Management Group), Department of Mathematics and Applied Mathematics, University of Cape Town, Rondebosch, 7701, South Africa.

¹ A hinge point for CTS should likely be higher than the 8th lowest SSB (de Moor 2025b). Considering the change in medians as the hinge point increased did not reveal any obvious candidate hinge points. If this method is preferred, alternative points between the 10th and 20th should be considered.

ordered historical years for CTS. There are 30 or 20 years above the hinge point for CTS and 29 years above the hinge point for WTS, as the three pulse years are excluded.

Projections assuming no future catch are run with:

i) No future recruitment variability, i.e.

$$a) \quad R_y^j = a^j.$$

$$b) \quad R_y^j = \begin{cases} a^j & \sum_c SSB_{c,y}^j > b^j \\ \frac{a^j \sum_c SSB_{c,y}^j}{b^j} & \sum_c SSB_{c,y}^j \leq b^j \end{cases}.$$

ii) Future recruitment variability, i.e.

$$a) \quad R_y^j = a^j e^{\varepsilon_y^j \sigma_r^j}, \text{ with } \varepsilon_y^j \sim N(0,1).$$

$$b) \quad R_y^j = \begin{cases} a^j e^{\varepsilon_y^j \sigma_r^j} & \sum_c SSB_{c,y}^j > b^j \\ \frac{a^j \sum_c SSB_{c,y}^j}{b^j} e^{\varepsilon_y^j \sigma_r^j} & \sum_c SSB_{c,y}^j \leq b^j \end{cases}, \text{ with } \varepsilon_y^j \sim N(0,1).$$

Option (b-ii) is additionally simulated with future catch according to the candidate Management Procedure described in de Moor (2025c).

Results and Discussion

Figure 1 shows the stock recruitment ‘relationships’ estimated for the first 60 draws from the posterior distribution for this assessment. For (a), the median and 90%ile for a^{CTS} is 13.5 [9.6,19.0], for a^{WTS} is 68.7 [51.3,92.3], for σ_r^{CTS} is 2.00 [1.69,2.30] and for σ_r^{WTS} is 1.10 [0.84,1.37]. For (b) the median and 90%ile for a^{WTS} is 86.4 [63.3,114.5], for b^{WTS} is 345 [191,564] and for σ_r^{WTS} is 0.89 [0.66,1.16]. Assuming a hinge point of the 10th lowest SSB, the median and 90%ile for a^{CTS} is 16.3 [10.8,24.1], for b^{CTS} is 117 [72,189] and for σ_r^{CTS} is 1.78 [1.37,2.17]. Assuming a hinge point of the 20th lowest SSB, the median and 90%ile for a^{CTS} is 19.5 [11.2,29.9], for b^{CTS} is 448 [320,613] and for σ_r^{CTS} is 1.66 [1.20,2.17].

Figures 2-5 shows the future recruitment generated from these ‘relationships’, both with and without inter-annual variability about these relationships. The Hockey-Stick ‘like’ relationship with the lower hinge points projects similar ranges to that of the median, with the projected recruitment for WTS being higher for (b) than (a) given the higher estimated a^{WTS} . Given the lack of feedback for the median ‘relationship’, there is no decrease over time in the lower percentiles (Figure 2). This decrease is seen for the Hockey-Stick ‘like’ relationship once lower recruitments are simulated resulting in lower spawner biomass and associated feedback from the decreasing limb of the relationship (Figure 3). The Hockey-Stick ‘like’ relationship with the higher hinge point for CTS has more than a 50% chance of current spawner biomass being below the hinge point and thus falling on the decreasing limb of the relationship. This is negated to a large degree once annual deviations from the relationship are included (Figure 4).

Figure 5 shows a scatter plot of historical and future spawner biomass – recruitment pairs. These show the impact of the upper bound of the currently assumed prior distribution if historical recruitment is estimated independently. (This impact does not occur when a stock recruitment relationship is estimated while fitting the assessment to the data (de Moor 2025d)). If this method is preferred, alternative prior distributions will need to be investigated for R_y^j . These figures also show that the standard deviation

generated using the method described above results in a substantially wider distribution of future recruitments (with subsequent impact on biomass) than that estimated historically.

Figures 6 and 7 show the historical and future spawner biomass while Figures 8 and 9 show the historical and future total November biomass. These indicate that one of the primary causes of wide ranges in the projected biomass is due to the inter-annual variability about the assumed stock recruitment 'relationship'.

While not inconsistent with inter-annual changes in other historical years, there are some substantial changes in spawner/total biomass between 2024 and 2025. The increase in west coast November biomass is likely due to the relatively high CTS recruitment estimated for October 2024, with only some of these being mature by October 2025 (CTS recruitment in October 2023 was estimated to be relatively low likely explaining the decrease in CTS spawner biomass).

References

- de Moor CL. 2025a. Considering only variability in annual recruitments in the sardine assessment. DFFE: Branch Fisheries Document FISHERIES/2025/SEP/SWG-PEL/50.
- de Moor CL. 2025b. Further Responses to Requests from the Technical Task Group. DFFE: Branch Fisheries Document FISHERIES/2025/OCT/SWG-PEL/56.
- de Moor CL. 2025c. Scoping MSE for South African sardine. International Stock Assessment Workshop Document IWS/2025/Sardine/BG5.
- de Moor CL. 2025d. Projections of the sardine population using an operating model with parametric stock-recruitment relationships estimated during conditioning. International Stock Assessment Workshop Document IWS/2025/Sardine/P4.

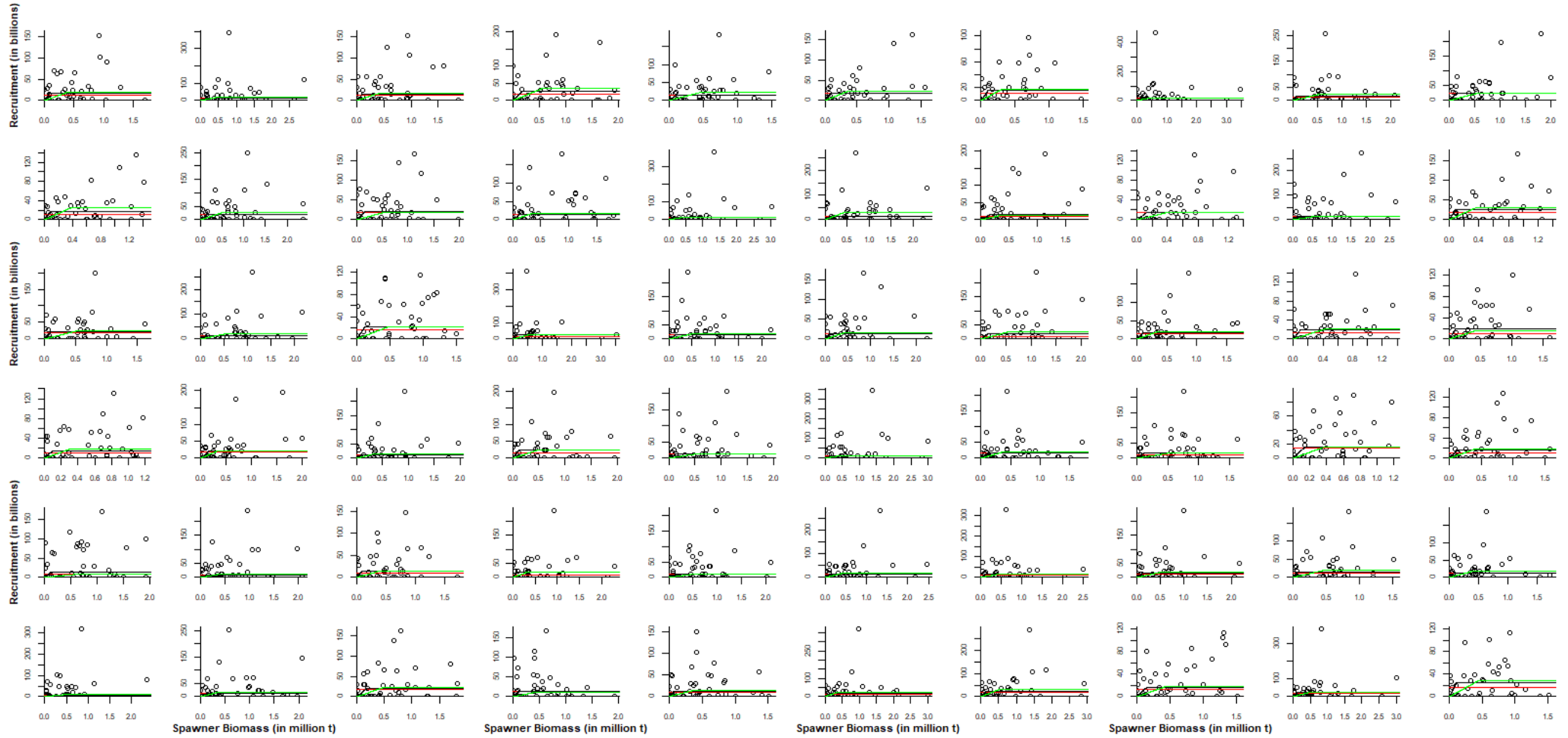


Figure 1a. Estimates of 1984-2023 CTS October recruitment against spawner biomass for 60 different replicates (individual plots), together with the estimated median (red) recruitment and the estimated Hockey Stick-type stock recruitment relationship, assuming the hinge point corresponds to the 10th (black) and 20th (green) lowest spawner biomass. The final year (2024) is excluded from these relationships and not plotted here.

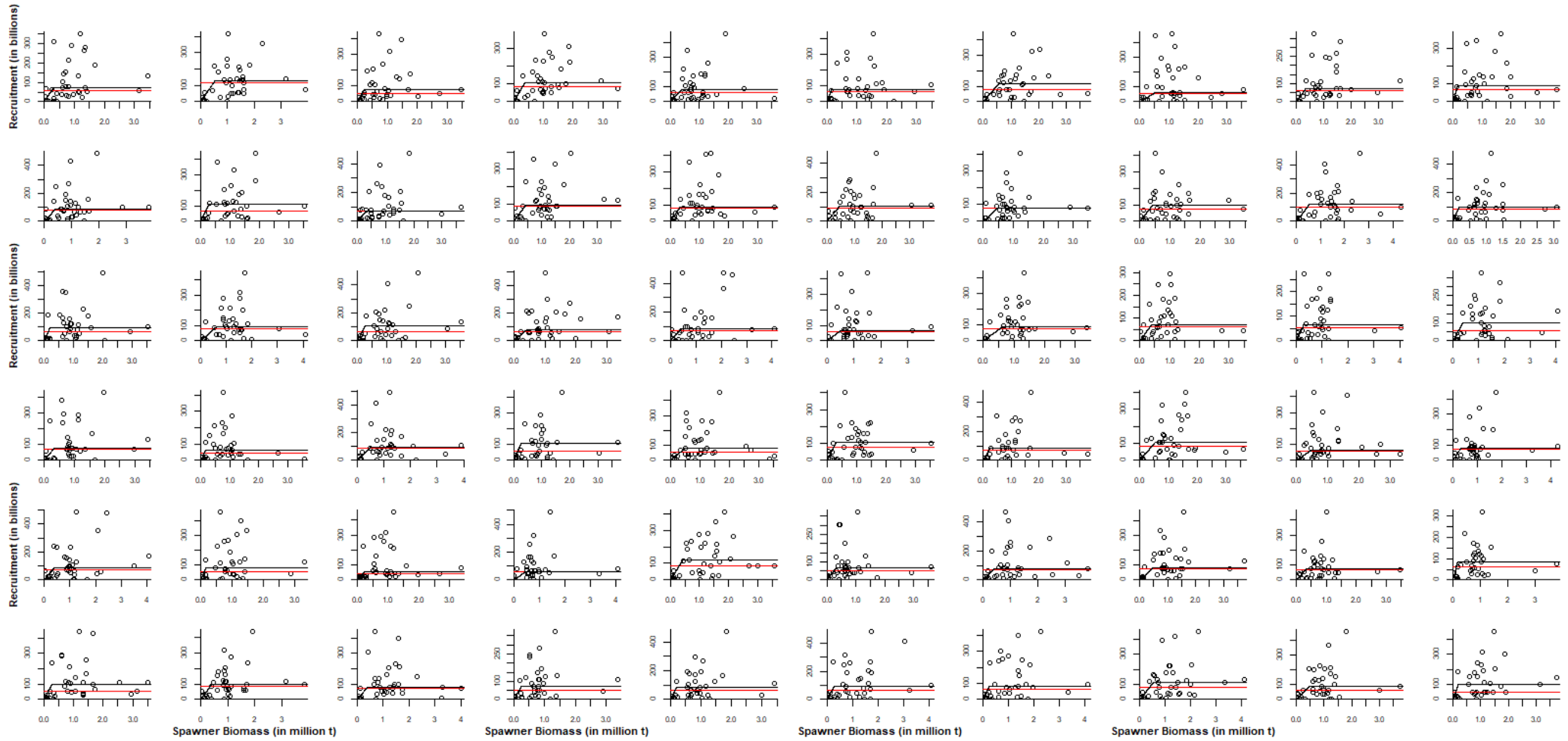


Figure 1b. Estimates of 1984-2023 WTS October recruitment against spawner biomass for 60 different replicates (individual plots), together with the estimated median (red) recruitment and the estimated Hockey Stick-type stock recruitment relationship, assuming the hinge point corresponds to the 8th (black) lowest spawner biomass. The final year (2024) is excluded from these relationships and not plotted here.

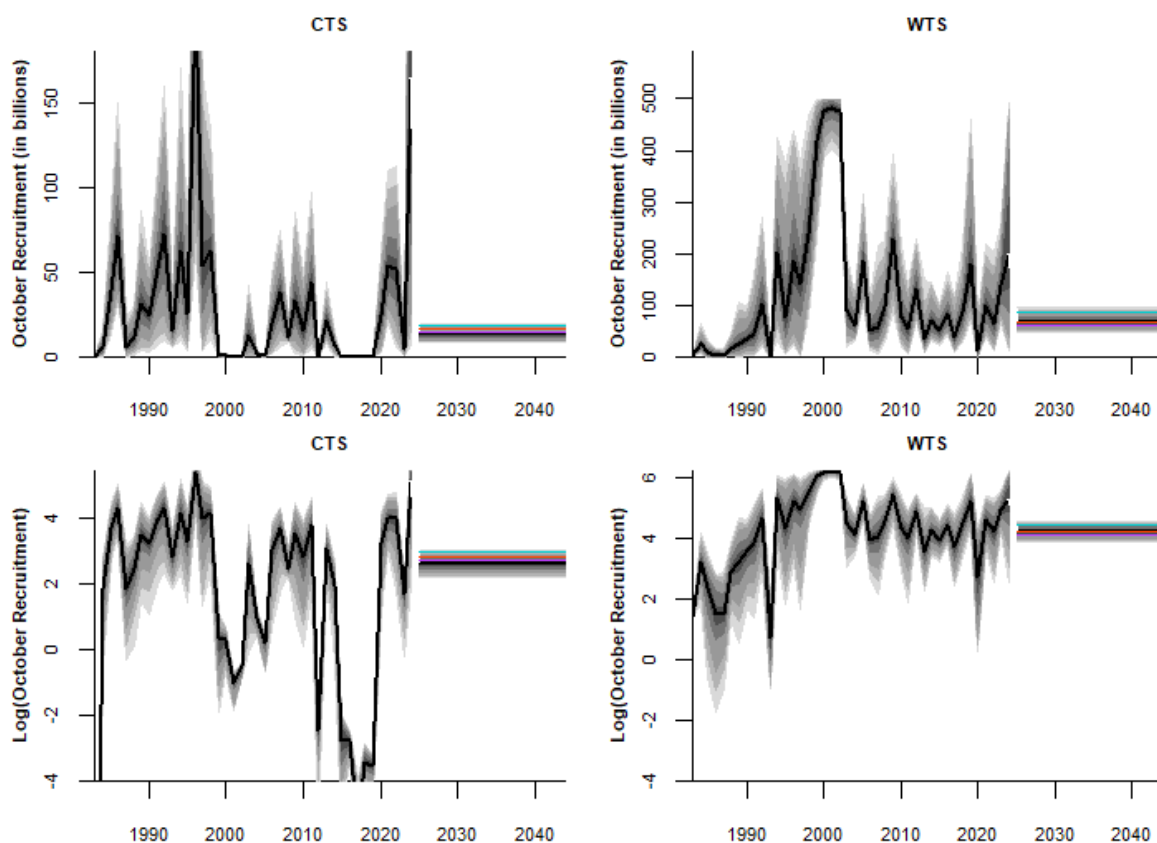


Figure 2a. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected October recruitment for CTS (left) and total WTS (right) under a no future catch scenario and (a-i) median recruitment with no future recruitment variability. Trajectories of three replicates are also shown in colour. The lower row is a repeat of the top, but with recruitment in log-space, in accordance with the error distribution assumed.

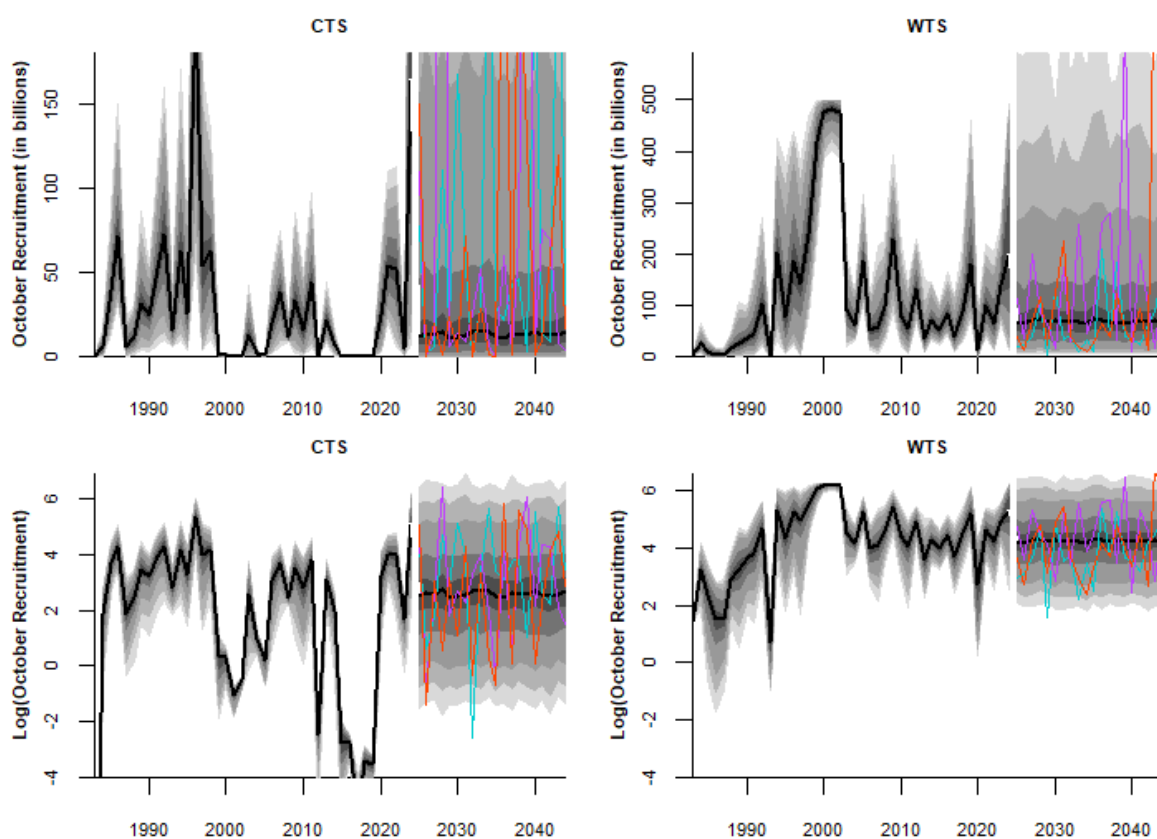


Figure 2b. As for Figure 2a, but with (a-ii) median recruitment with future recruitment variability.

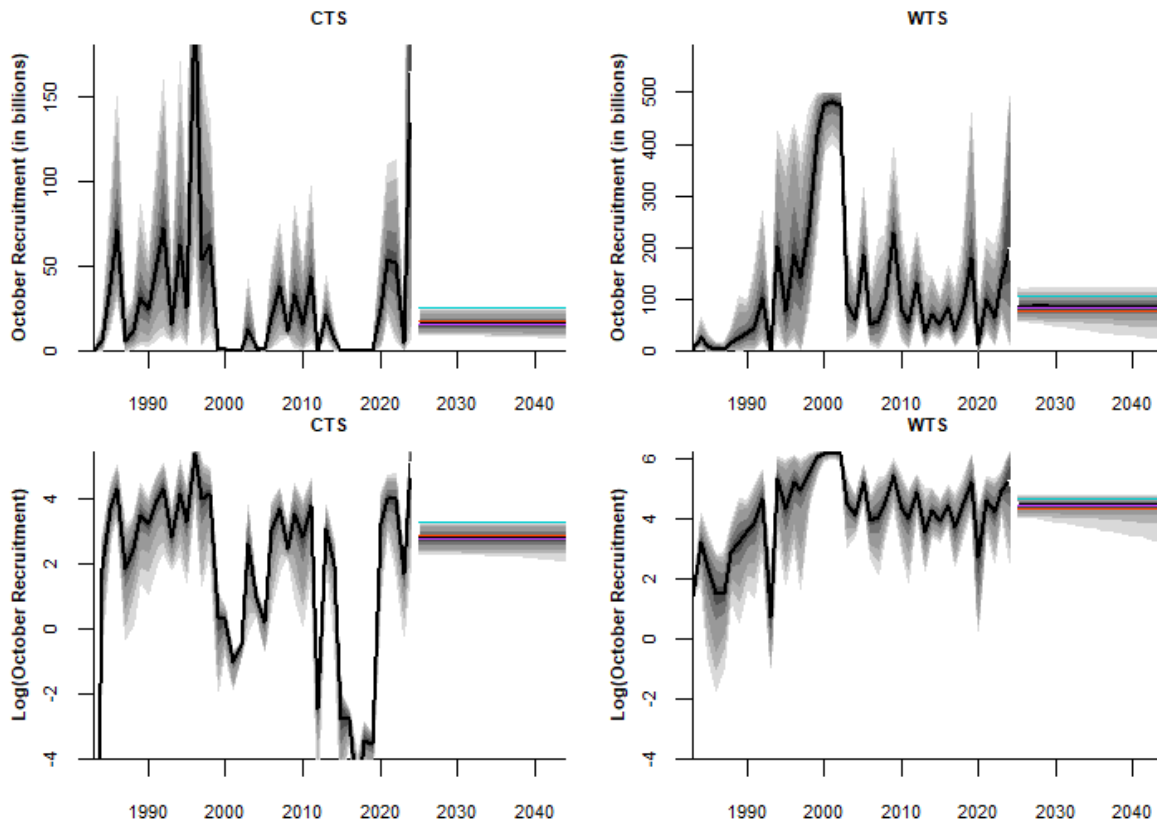


Figure 3a. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected October recruitment for CTS (left) and total WTS (right) under a no future catch scenario and (b-i) Hockey-Stick 'like' recruitment assuming the 10th lowest SSB for the hinge point for CTS, with no future recruitment variability. Trajectories of three replicates are also shown in colour. The lower row is a repeat of the top, but with recruitment in log-space, in accordance with the error distribution assumed.

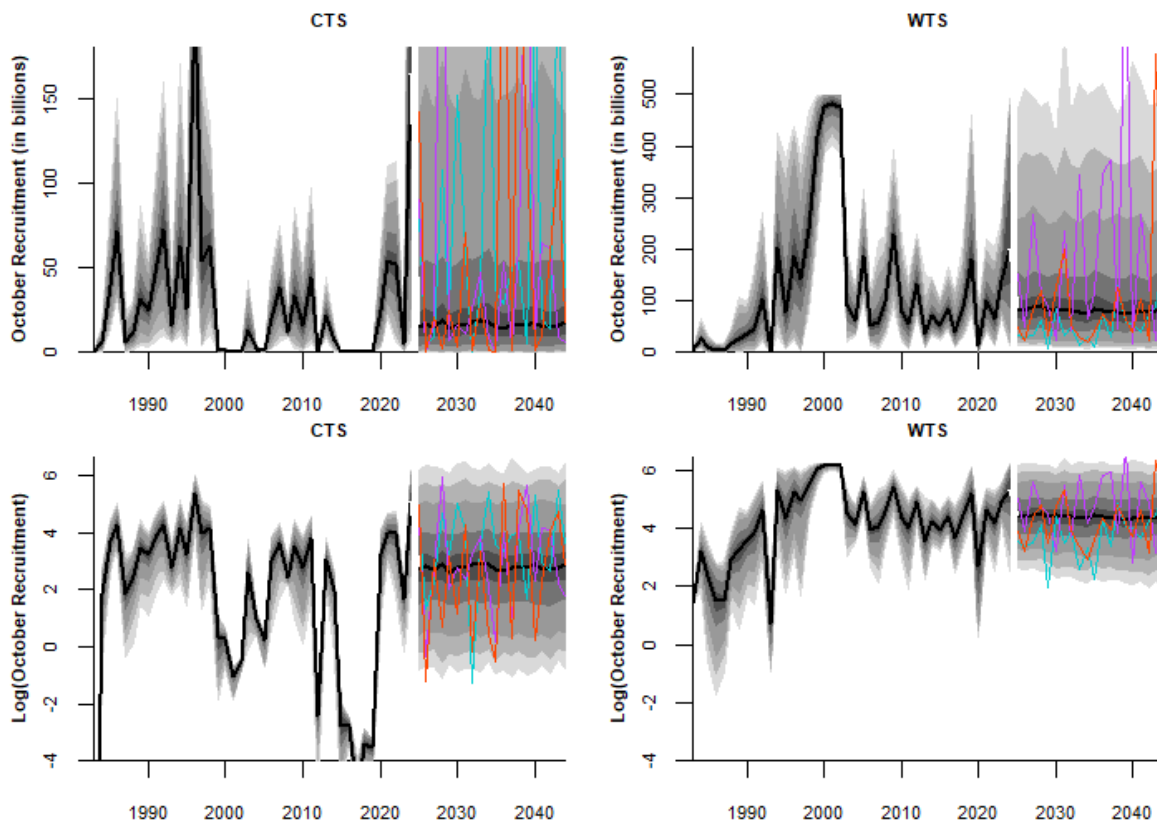


Figure 3b. As for Figure 3a, but with (b-ii) Hockey-Stick 'like' recruitment with future recruitment variability.

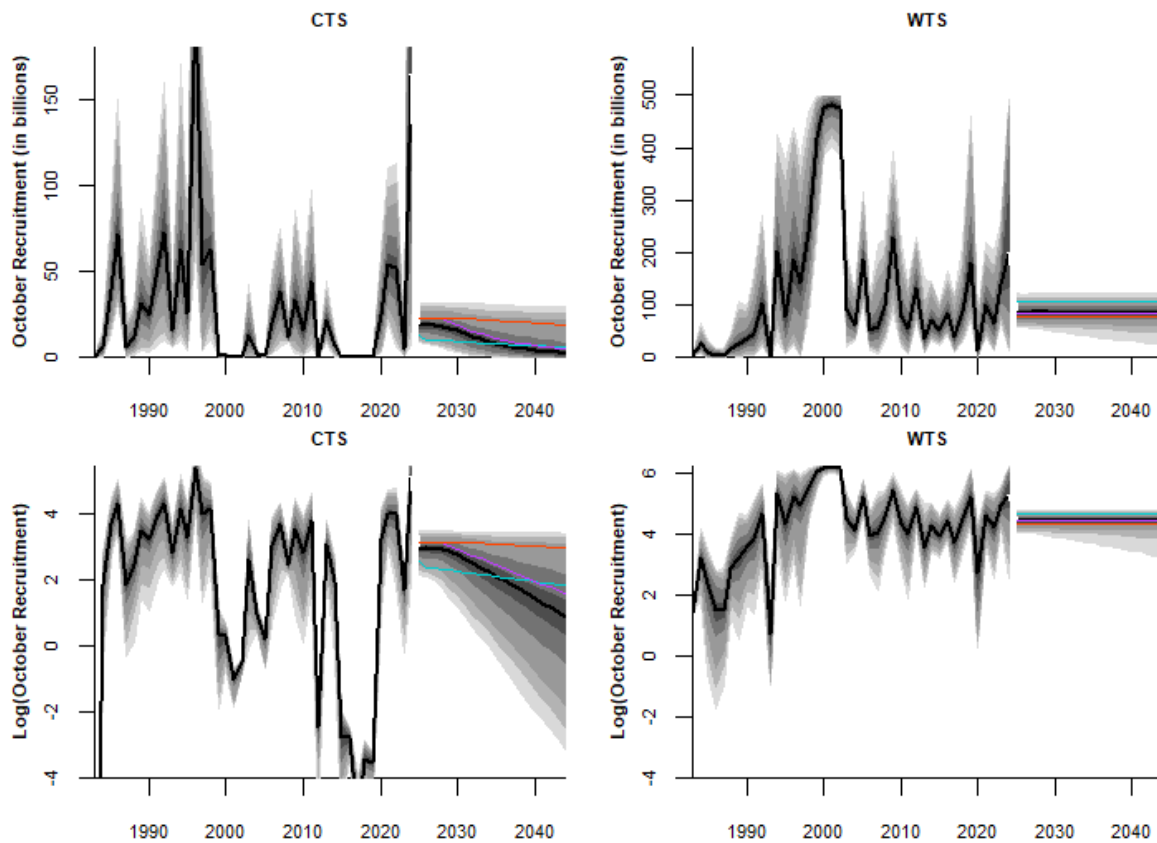


Figure 4a. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected October recruitment for CTS (left) and total WTS (right) under a no future catch scenario and (b-i) Hockey-Stick 'like' recruitment assuming the 20th lowest SSB for the hinge point for CTS, with no future recruitment variability. Trajectories of three replicates are also shown in colour. The lower row is a repeat of the top, but with recruitment in log-space, in accordance with the error distribution assumed.

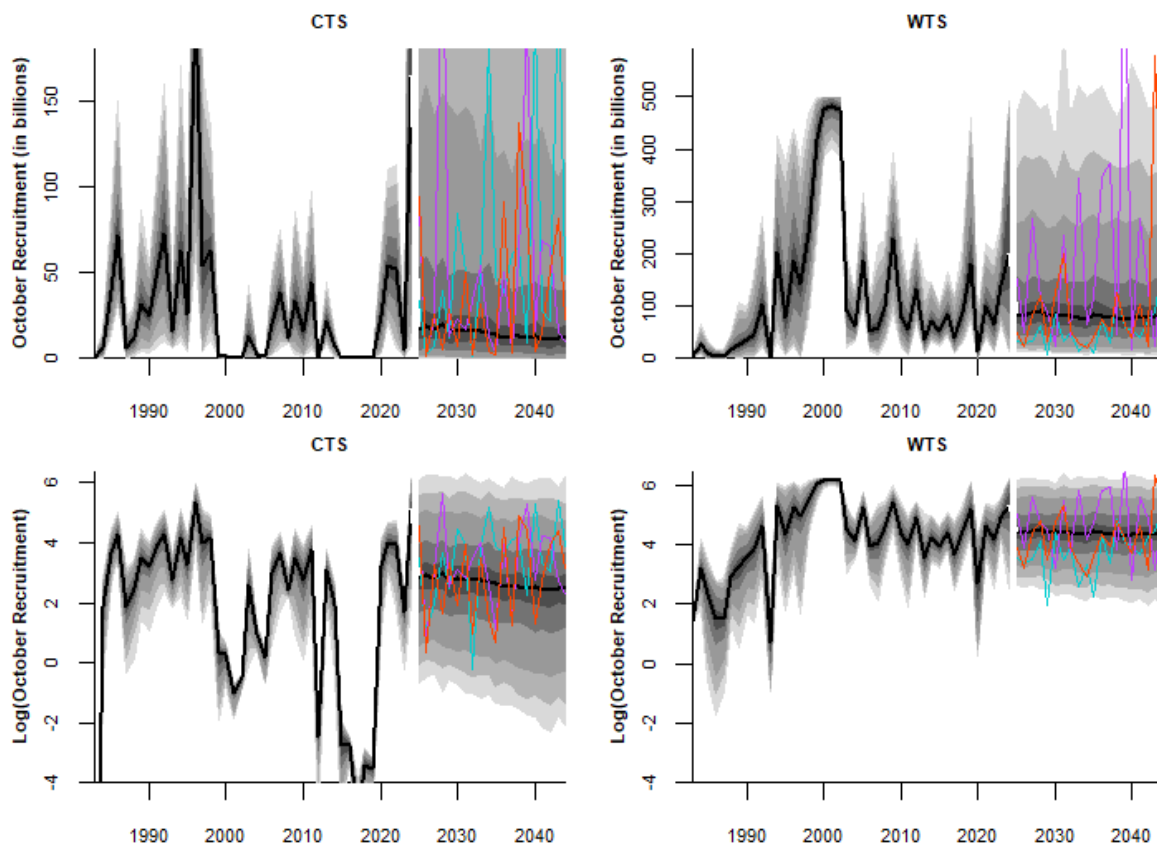


Figure 4b. As for Figure 4a, but with (b-ii) Hockey-Stick 'like' recruitment with future recruitment variability.

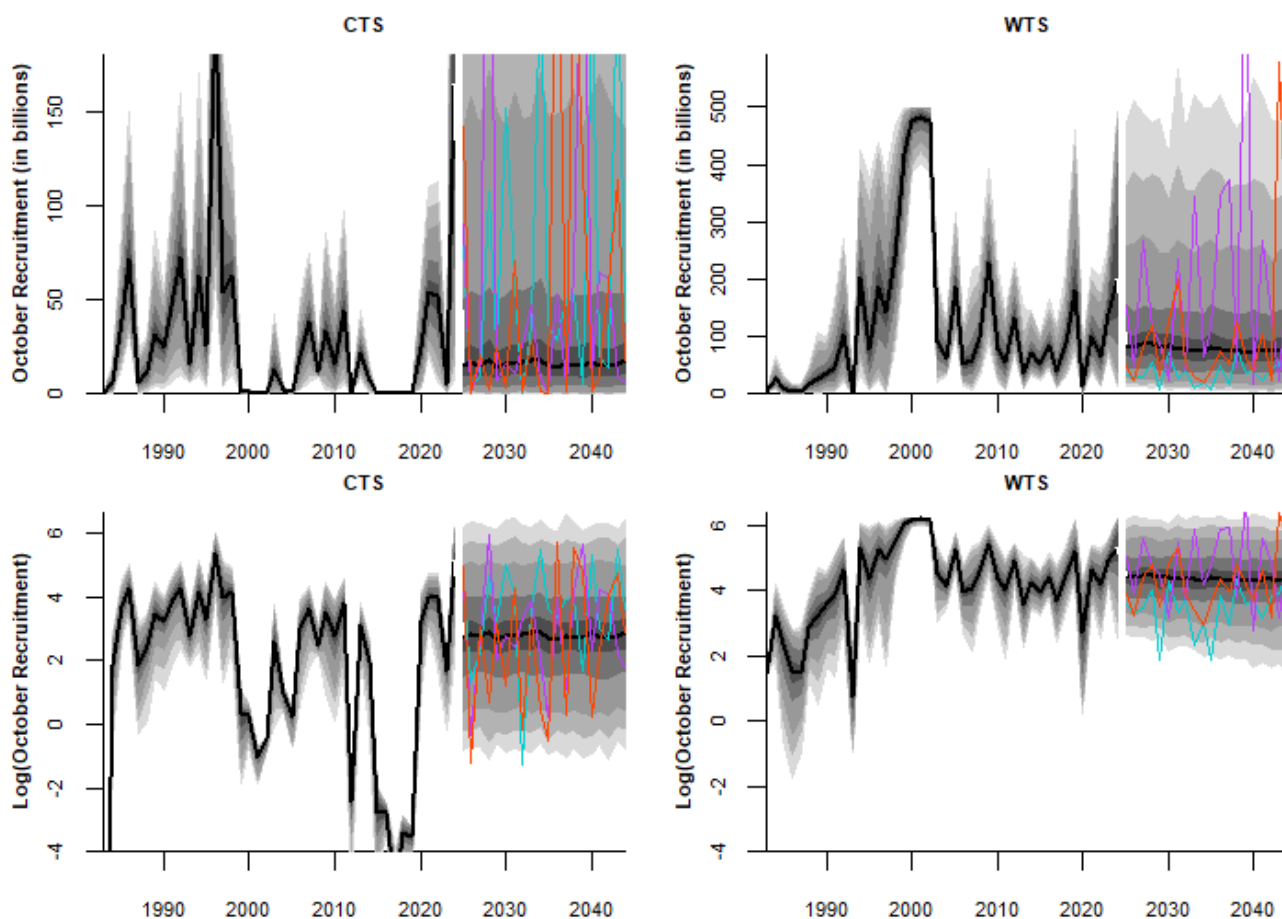


Figure 4c. As for Figure 4b, but with future catch.

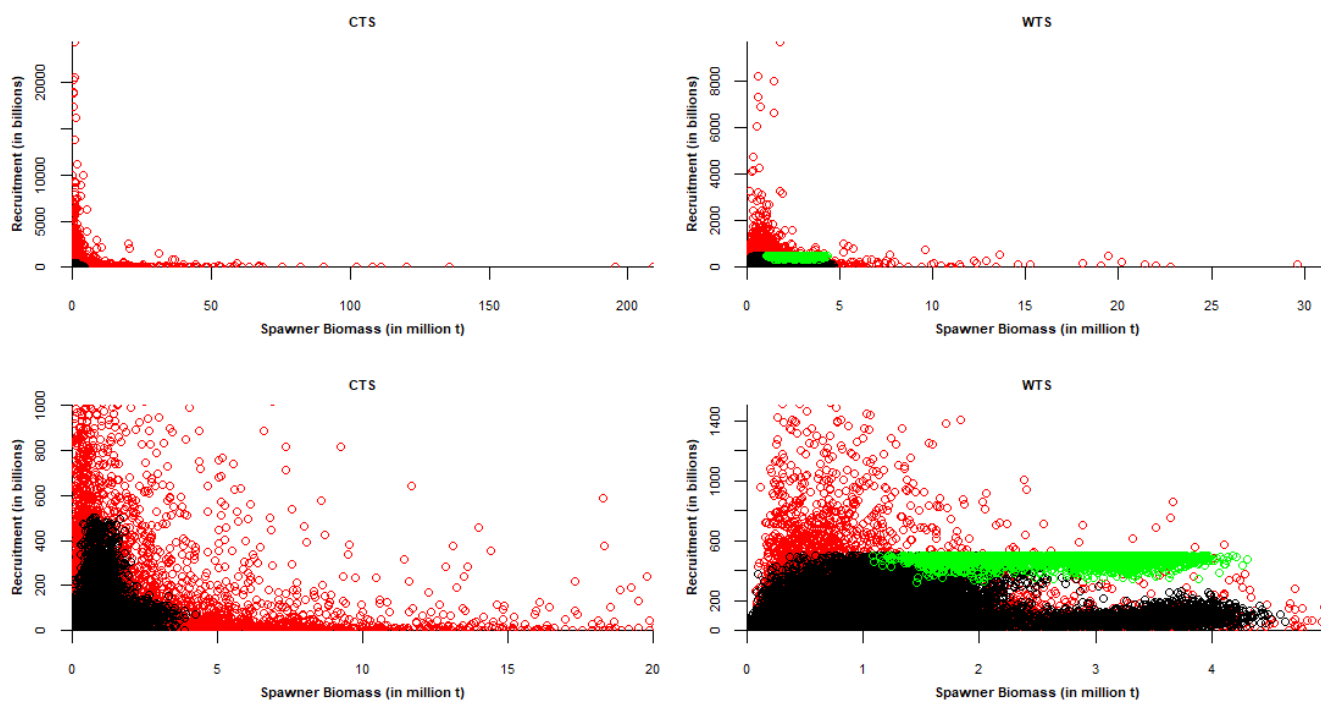


Figure 5a. Pairs of October recruitment and spawner biomass for CTS (left) and WTS (right) under a no future catch scenario for (a-ii) median recruitment with future recruitment variability. The black circles are historical estimates, with green showing the three pulse years. The red circles are future estimates. The lower plot is a repeat of the top on a different scale.

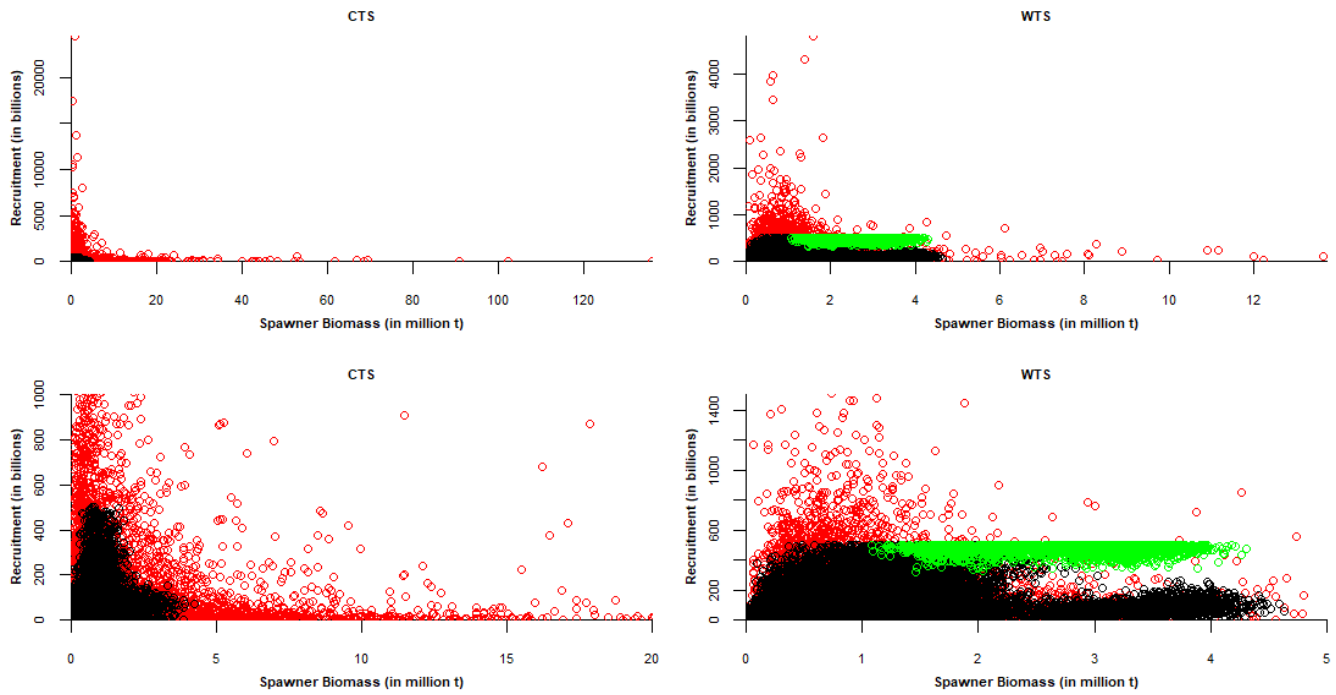


Figure 5b. Pairs of October recruitment and spawner biomass for CTS (left) and WTS (right) under a no future catch scenario for (b-ii) Hockey Stick type recruitment with future recruitment variability. The black circles are historical estimates, with green showing the three pulse years. The red circles are future estimates. The lower plot is a repeat of the top on a different scale.

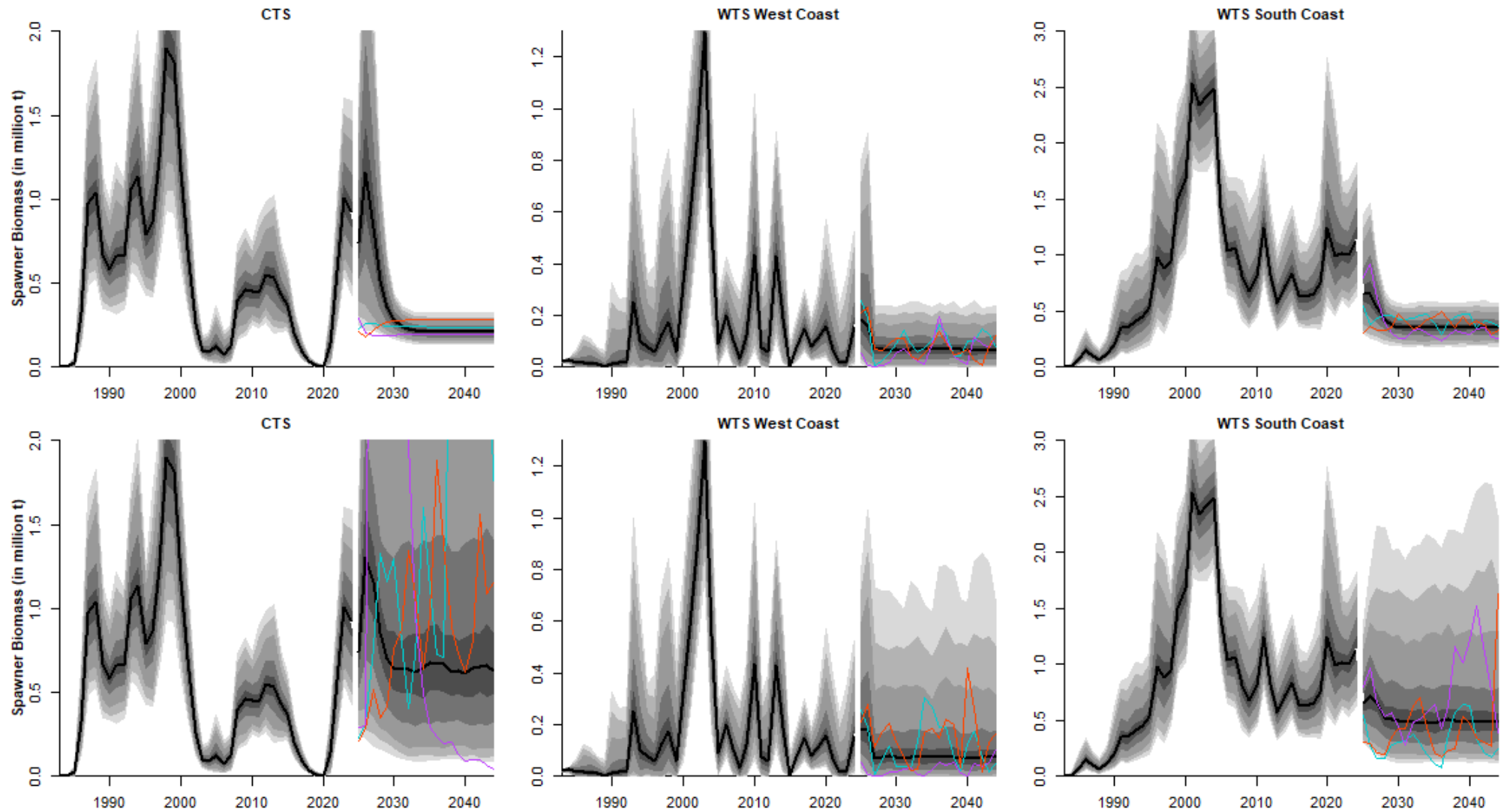


Figure 6. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected spawner biomass for CTS (left) and WTS on the west (middle) and south (right) coasts under a no future catch scenario and (a) median recruitment with (i) no recruitment variability (top row) and (ii) with future recruitment variability (lower row). Trajectories of three replicates are also shown in colour.

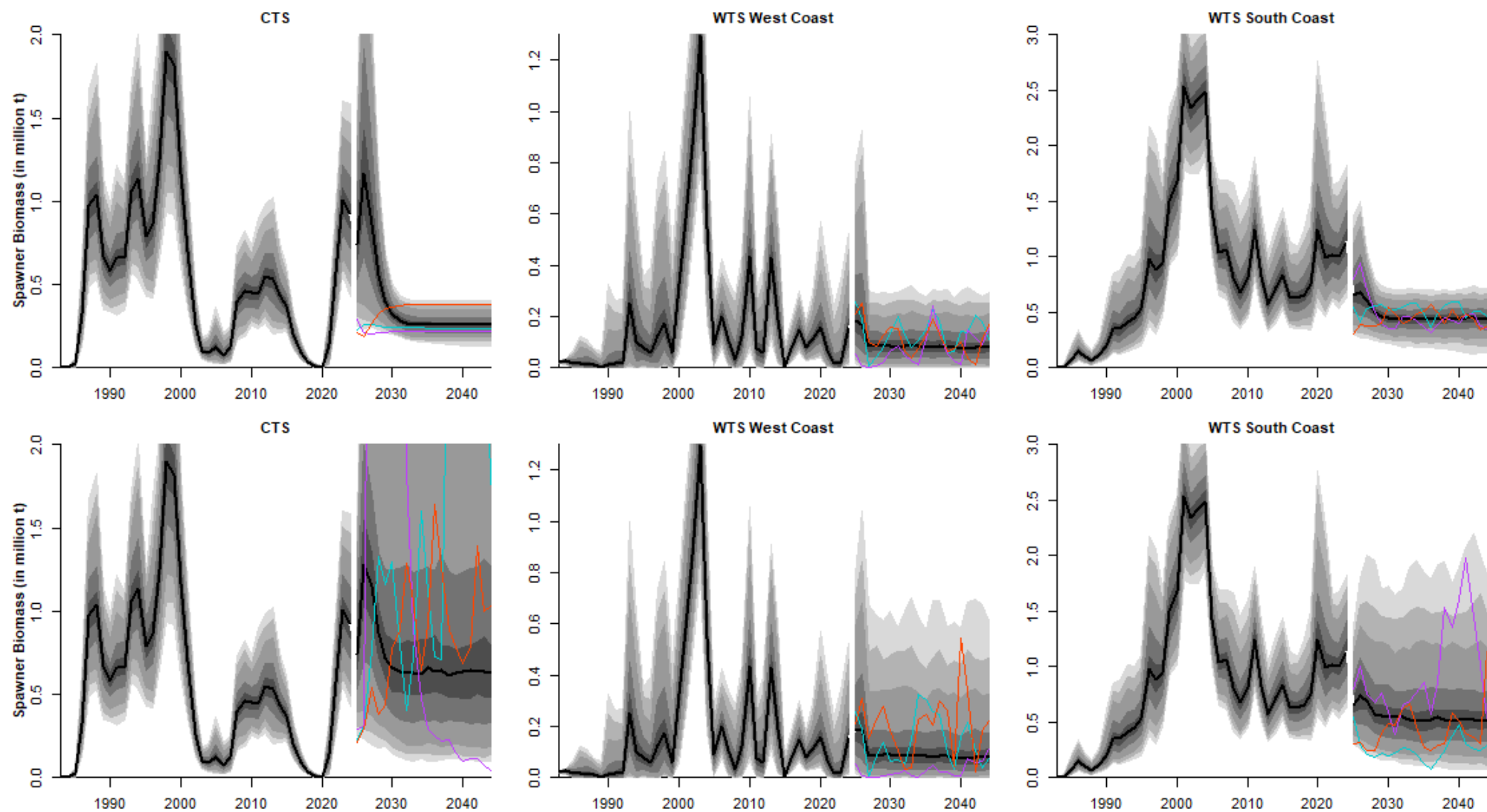


Figure 7. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected spawner biomass for CTS (left) and WTS on the west (middle) and south (right) coasts under (b) Hockey-Stick 'like' recruitment with (i) no recruitment variability and no future catch (top row), with (ii) future recruitment variability and no future catch (middle row), and with (ii) future variability and future catch (lower row). Trajectories of three replicates are also shown in colour.

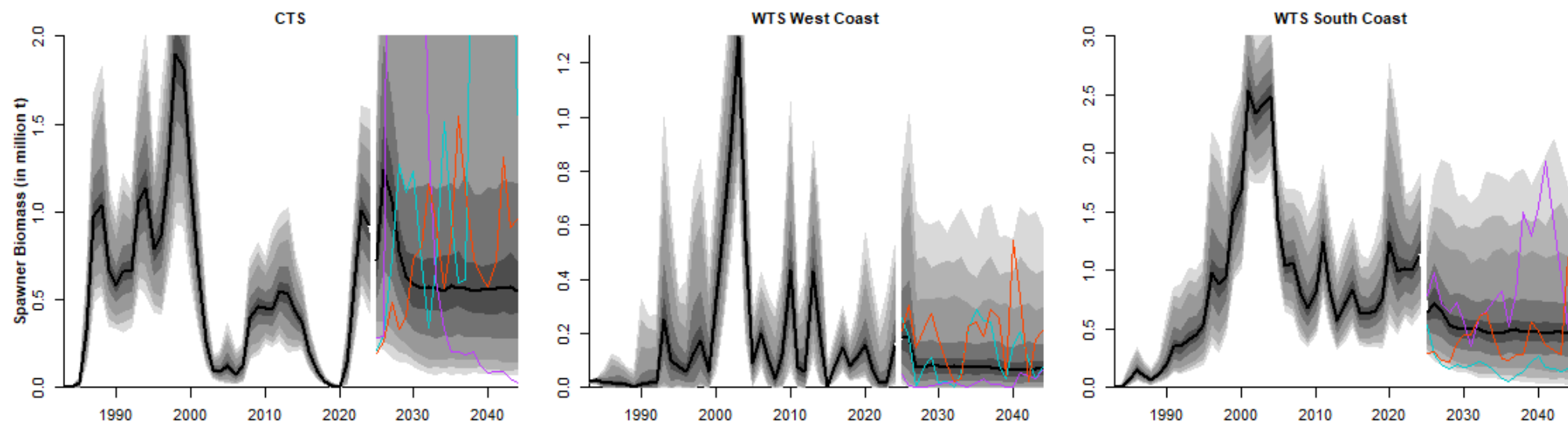


Figure 7 (continued).

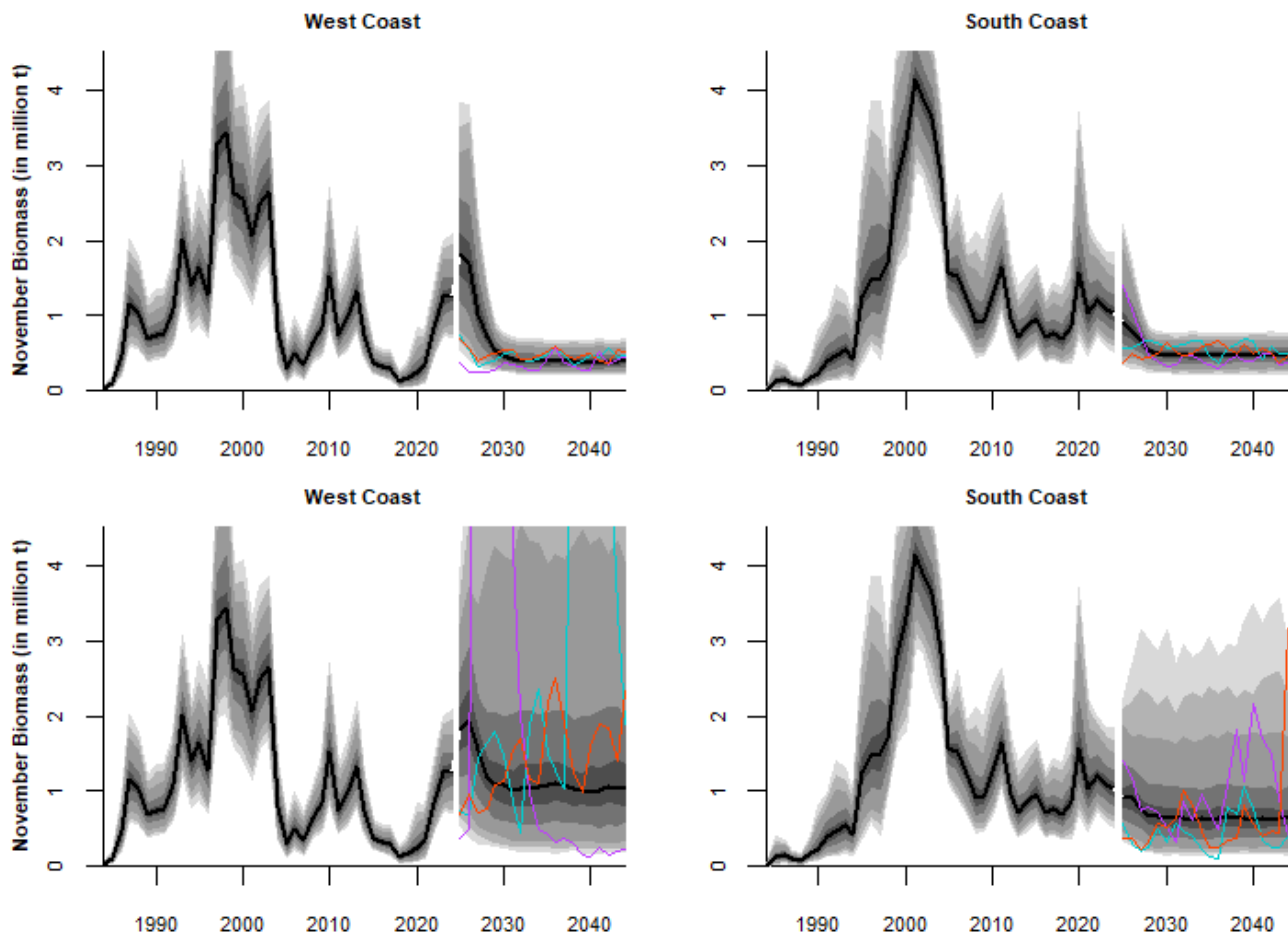


Figure 8. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected total November biomass for the west and south coasts under a no future catch scenario and (a) median recruitment with (i) no recruitment variability (top row) and (ii) with future recruitment variability (lower row). Trajectories of three replicates are also shown in colour.

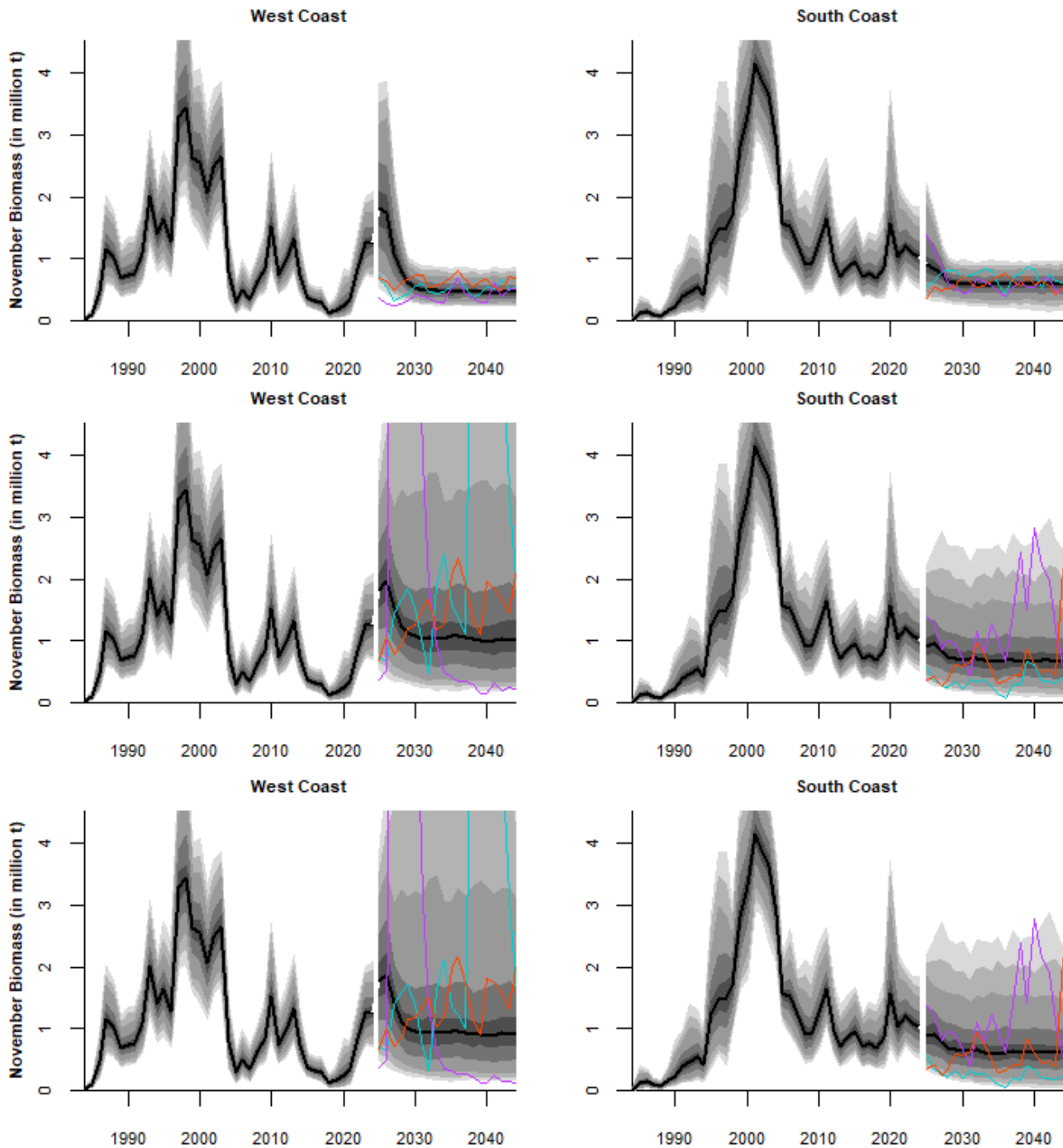


Figure 9. Median (solid line), 20, 50, 80, 90 and 95% probability intervals (in increasingly lighter shades of grey) of historically estimated and projected total November biomass for the west and south coasts under a no future catch scenario and (b) Hockey-Stick 'like' recruitment with the hinge point for CTS fixed at the 10th lowest SSB with (i) no recruitment variability and no future catch (top row), with (ii) future recruitment variability and no future catch (middle row), and with (ii) future variability and future catch (lower row). Trajectories of three replicates are also shown in colour.